

finding the bit string that maximizes the "One-Max" function (count of 1s): $p=4, r=0.5, m=0.01$

let $P = \{h_1: 10001, h_2: 01010, h_3: 11000, h_4: 00010\}$

→ Initialization done

→ evaluation fitness function: -

$h_1: 2, h_2: 2, h_3: 2, h_4: 1$

$$\text{Total} = 2+2+2+1 = 7$$

→ selection: - It is done using fitness proportionate selection

$$\text{Probabilities: } h_1 = \frac{2}{7}, h_2 = \frac{2}{7}, h_3 = \frac{2}{7}, h_4 = \frac{1}{7}$$

$$r \cdot p \Rightarrow (0.5)(4) = 2$$

∴ 2 individuals undergo crossover.

$$(1-r)p \Rightarrow (1-0.5)(4) = 2$$

∴ 2 individuals are directly selected (or) copied to next population. let h_1, h_2, h_3

$$\rightarrow \text{Crossover: - } \frac{r \cdot p}{2} = \frac{(0.5)(4)}{2} \quad (50\% \text{ crossover})$$

50% of 4 = 2 offsprings

let do crossover for h_1 & h_3

$$\begin{array}{rcl} 10001 & \rightarrow & 10000 \quad \text{--- ①} \\ 11000 & & 11001 \quad \text{--- ②} \end{array}$$

→ Mutation: - flip random members of P_1

$$\Rightarrow 0.01 \times 4 = 0.04$$

On an avg 0.04 bits mutate per generation.

lets mutate ②, $11001 \rightarrow 11101$

New population : 10000, 11101, 10001, 01010

Fitness function :- $h_1=1$ $h_2=4$ $h_3=2$ $h_4=2$

$$\text{Total} = 1+4+2+2 = 9$$

Selection : $h_1=1/9$ $h_2=4/9$ $h_3=2/9$ $h_4=2/9$

$$r \cdot p = (0.5)(4) = 2 \quad (1-r)p = (0.5)(4) = 2$$

let h_2, h_3 & h_4 are selected.

crossover :- 50% $4 = 2$ let h_2 & h_4

$$\begin{array}{r|l} 11101 & \text{--- ③} \\ 01010 & \text{--- ④} \end{array} \rightarrow \begin{array}{r|l} 11110 & \text{--- ③} \\ 01001 & \text{--- ④} \end{array}$$

Mutation :- $m \cdot p = 0.01 \times 4 = 0.04$

lets mutate ③ $11110 \rightarrow 11111$

New population :- 11111, 01001, 11101, 10001

Optimality condition reached. //

Items : 1 2 3 4 $W=50$

values 100 150 100 100 $r=0.8$

weights 20 30 10 20 $P=5$

Population:	Chromosome	Probability	fitness
	1011	6/25	300
	1100	1/5	250
	0101	1/5	250
	1001	4/25	200
	0110	1/5	250
			1250

Selection :- $(r \cdot P) = (0.8)(5) = 4 \rightarrow \text{Crossover}$

$(1-r)P = (0.2)(5) = 1 \rightarrow \text{Direct}$

let h_1, h_2, h_3, h_5 are selected.

crossover:- $\begin{matrix} 1011 \\ 1100 \end{matrix} \rightarrow \begin{matrix} 1000 \rightarrow 100 \checkmark \\ 1111 \rightarrow \text{invalid} \end{matrix}$

$\begin{matrix} 01101 \\ 01110 \end{matrix} \rightarrow \begin{matrix} 0110 \rightarrow 250 \checkmark \\ 0101 \rightarrow 250 \checkmark \end{matrix}$

Mutation:- $m.p = 0.01 \times 5 = 0.05$

let $1000 \rightarrow \cancel{100} \rightarrow \cancel{250} / 1010 \rightarrow 200$

New population:- $1011, 1100, 0101, 0110, \cancel{1010}$

$300 + 250 + 250 + 250 + 200$

$= 1250$

probabilities:- $\frac{6}{25}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{4}{25}$

crossover:- $\begin{matrix} \cancel{1011} \\ \cancel{1100} \end{matrix} \rightarrow \begin{matrix} 1011 \rightarrow 1010 \rightarrow 200 \checkmark \\ 0110 \rightarrow 0111 \rightarrow \text{invalid} \end{matrix}$

$\begin{matrix} 1100 \\ 01101 \end{matrix} \rightarrow \begin{matrix} 1101 \rightarrow \text{invalid} \\ 0100 \rightarrow 150 \checkmark \end{matrix}$

Mutation:- $0100 \rightarrow 0110 (250) \checkmark$

New population:- 1011, 1100, 0110, 1010, 0101

300 is maximum possible value, as we checked earlier, GA has already found optimal sol.

(u)

$$P(w|D) = \frac{P(D|w)P(w)}{P(D)}$$

Ex- student	Hours studied (x)	Result (y)
1	1	0
2	2	0
3	5	1
4	6	1

Compare the hypothesis using "weight" parameter

$$\text{let } w_1 = 0.1 \quad w_2 = 1.0$$

$$P(w_1) = 0.5 \quad P(w_2) = 0.5$$

$$P(y|p) = p^y (1-p)^{1-y} \quad (\text{Binomial})$$

→ In discriminative model $\hat{y}_i = \sigma(wx_i)$ the likelihood of an observation

$$\therefore P(y | x_i, w) = \hat{y}_i^{y_i} (1 - \hat{y}_i)^{1-y_i} \quad p = \sigma(wx_i) \quad y = y_i$$

→ Discriminative likelihood for dataset is:

$$P(D|w) = \prod_{i=1}^n \sigma(wx_i)^{y_i} (1 - \sigma(wx_i))^{1-y_i}$$

∴ likelihood for $w_1 = 0.1$:

$$P(D|w_1) = \sigma(0.5) \cdot \sigma(0.6) (1 - \sigma(0.1)) \cdot (1 - \sigma(0.2))$$

$$(x=1, y=0) \text{ \& } w_1 = 0.1 \quad \sigma(wx_i)^y = \sigma(0.1 \times 1)^0 = (0.525)^0 = 1$$

$$(1 - \sigma(0.1))^{1-y} = 1 - \sigma(0.1) = 0.475$$

$$(x=2, y=0) \therefore w_1 = 0.1 \quad \sigma(0.2)^0 = 0.55 \quad (1 - \sigma(0.2))^{1-y} = 0.45$$

$$(x=5, y=1) \therefore w_1 = 0.1 \quad \sigma(0.5)^1 = 0.622 \quad (1 - \sigma(0.5))^{1-y} = 0$$

$$(1 - \sigma(0.5))^{1-y} = 1$$

$$(x=6, y=1) \quad w_1 = 0.1 \quad \sigma(0.6)^1 = 0.646 \quad (1 - \sigma(0.6))^{1-y} = 1$$

$$\therefore P(D|w_1) = (1 - \sigma(0.1)) (1 - \sigma(0.2)) \sigma(0.5) \sigma(0.6)$$

$$= 0.475 \times 0.450 \times 0.622 \times 0.646 = 0.0859$$

$$P(D|\omega_2) = (1 - \sigma(1.0)) (1 - \sigma(2.0)) \sigma(5.0) \sigma(6.0)$$

$$= 0.0317$$

$$\therefore \text{posterior } P(\omega_j|D) = \frac{P(D|\omega_j) P(\omega_j)}{\sum_k P(D|\omega_k) P(\omega_k)}$$

$$\sum_k P(D|\omega_k) P(\omega_k) = P(D|\omega_1) P(\omega_1) + P(D|\omega_2) P(\omega_2)$$

The weak correlation model fits in small dataset better.

$$= 0.0859 \times 0.5 + 0.0317 \times 0.5$$

$$= 0.0588$$

$$P(\omega_1|D) = \frac{P(D|\omega_1) \cdot P(\omega_1)}{\sum_k P(D|\omega_k) P(\omega_k)} = \frac{0.0859 \times 0.5}{0.0588} = 0.7311$$

... regression: